

第 2 课 ROS 系统安装和环境搭建

1.版本选择

不同的 Ubuntu 安装的 ROS 版本不同，其中 Ubuntu18.04 的 ROS 对应版本为 Melodic。

Ubuntu版本	ROS版本
Ubuntu16.04	Kinetic
Ubuntu18.04	Melodic
Ubuntu20.04	Noetic

2.检查 Ubuntu 的软件和更新源

- 1) 找到系统中的“软件和更新”。



- 2) 确保勾选上互联网下载的选项，下载源更换成国内的，以阿里云为例。



3.设置 ROS 的下载源

3.1 设置国内下载源

打开命令行终端，输入指令：

```
“sudo sh -c ‘. /etc/lsb-release && echo “deb http://mirrors.ustc.edu.cn/ros/ubuntu/`lsb_release -cs` main” > /etc/apt/sources.list.d/ros-latest.list’”，按下回车，接着输入虚拟机的密码“ubuntu”（此密码为用户自行设置的，我们这里的密码为 ubuntu）。
```

```
File Edit View Search Terminal Help
ubuntu@ubuntu-virtual-machine:~$ sudo sh -c '. /etc/lsb-release && echo "deb ht
tp://mirrors.ustc.edu.cn/ros/ubuntu/ `lsb_release -cs` main" > /etc/apt/sources
.list.d/ros-latest.list'
[sudo] password for ubuntu:
ubuntu@ubuntu-virtual-machine:~$
```

3.2 设置公匙

输入指令“`sudo apt-key adv --keyserver 'hkp://keyserver.ubuntu.com:80' --recv-key C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654`”按下回车，如下图所示则设置成功。

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt-key adv --keyserver 'hkp://keyserver.ubuntu.com:80' --recv-key C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
Executing: /tmp/apt-key-gpghome.JqM65juhYg/gpg.1.sh --keyserver hkp://keyserver.ubuntu.com:80 --recv-key C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
gpg: key F42ED6FBAB17C654: "Open Robotics <info@osrfoundation.org>" 1 new signature
gpg: Total number processed: 1
gpg:      new signatures: 1
ubuntu@ubuntu-virtual-machine:~$ sudo apt update
```

3.3 更新软件包

输入指令“`sudo apt update`”。

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt update
```

4. 安装 ROS

打开命令行终端，输入指令“`sudo apt install ros-melodic-desktop-full`”，等待下载安装完成，根据网络状况不同，安装过程大概要 10 分钟。

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt install ros-melodic-desktop-full
```

5. 设置环境变量

打开命令行终端，依次输入指令“`echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc`”和“`source ~/.bashrc`”，完成环境变量配置。

```
ubuntu@ubuntu-virtual-machine:~$ echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc
ubuntu@ubuntu-virtual-machine:~$ source ~/.bashrc
```

6. 安装 rosdep

输入指令“`sudo apt-get install python3-rosdep`”按下回车，安装 rosdep；出现安装确认提示，输入“Y”即可。

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt-get install python3-rosdep
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  python3-catkin-pkg-modules python3-docutils python3-pygments
  python3-pyparsing python3-roman python3-rosdep-modules
  python3-roscdistro-modules python3-rospkg-modules
Suggested packages:
  fonts-linuxlibertine | ttf-linux-libertine texlive-lang-french
  texlive-latex-base texlive-latex-recommended python-pyparsing-doc
The following packages will be REMOVED:
  python-rosdep
The following NEW packages will be installed:
  python3-catkin-pkg-modules python3-docutils python3-pygments
  python3-pyparsing python3-roman python3-rosdep python3-rosdep-modules
  python3-roscdistro-modules python3-rospkg-modules
0 upgraded, 9 newly installed, 1 to remove and 659 not upgraded.
Need to get 1,153 kB of archives.
After this operation, 6,290 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

7. 初始化

- 1) 输入指令“`cd /etc/ros/rosdep/sources.list.d`”进入“`sources.list.d`”文件夹。

```
ubuntu@ubuntu-virtual-machine:~$ cd /etc/ros/rosdep/sources.list.d
```

- 2) 输入指令“`sudo gedit 20-default.list`”，编辑此文件。

```
ubuntu@ubuntu-virtual-machine:/etc/ros/rosdep/sources.list.d$ sudo gedit 20-default.list
```

- 3) 将原文件中的内容替换为以下内容（**直接复制、粘贴，不需要换行，也不要手动输入**）。

```
# os-specific li# os-specific listings first yaml
https://gitee.com/wybro/roscdistro/raw/master/rosdep/osx-homebrew.yaml osx # generic yam
https://gitee.com/wybro/roscdistro/raw/master/rosdep/base.yaml yaml
https://gitee.com/wybro/roscdistro/raw/master/rosdep/python.yaml yaml
https://gitee.com/wybro/roscdistro/raw/master/rosdep/ruby.yaml gbpdistro
https://gitee.com/wybro/roscdistro/raw/master/releases/fuerte.yaml fuerte # newer distributions
(Groovy, Hydro, ...) must not be listed anymore, they are being fetched from the roscdistro
index.yaml instead
```

4) 修改过程中，终端会出现警告，属于正常现象，不必理会。

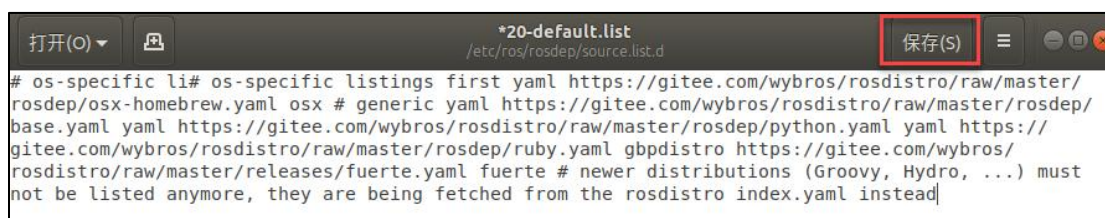
```
** (gedit:5086): WARNING **: 11:27:56.819: Set document metadata failed: Setting attribute metadata::gedit-position not supported
ubuntu@ubuntu-virtual-machine:/etc/ros/rosdep/sources.list.d$ sudo gedit 20-default.list

** (gedit:5129): WARNING **: 11:29:55.173: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported

** (gedit:5129): WARNING **: 11:29:55.174: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported

** (gedit:5129): WARNING **: 11:29:57.356: Set document metadata failed: Setting attribute metadata::gedit-position not supported
ubuntu@ubuntu-virtual-machine:/etc/ros/rosdep/sources.list.d$
```

5) 替换完成后，直接点击保存再关闭文件。



6) 输入指令“`cd /usr/lib/python2.7/dist-packages/rosdep2/`”,进入到“rosdep2”目录下。

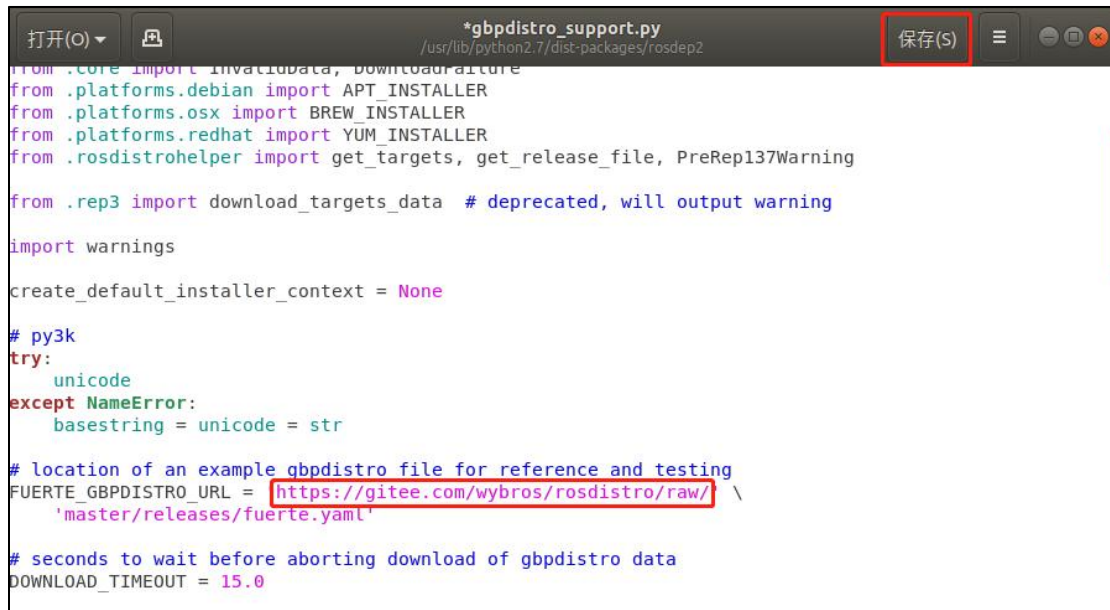
```
ubuntu@ubuntu-virtual-machine:/usr/lib/python2.7/dist-packages/rosdep2$ cd /usr/lib/python2.7/dist-packages/rosdep2/
```

7) 输入指令“`sudo gedit gbpdistro_support.py`”按下回车，若提示有输入虚拟机密码的提示，我们输入“ubuntu”（此密码为用户自行设置的，我们这里的密码为ubuntu），按下回车，即可打开“gbpdistro_support.py”文件，找到“FUERTE_GBPDISTRO_URL”代码。

```
ubuntu@ubuntu-virtual-machine:/usr/lib/python2.7/dist-packages/rosdep2$ sudo gedit gbpdistro_support.py

# location of an example gbpdistro file for reference and testing
FUERTE_GBPDISTRO_URL = 'https://raw.githubusercontent.com/ros/rosdistro/' \
    'master/releases/fuerte.yaml'
```

8) 将“`https://raw.githubusercontent.com/ros/rosdistro/`”修改为“`https://gitee.com/wybro/s/rosdistro/raw/`”，（只修改下图红框所示内容,后续内容无需更改）修改完成后点击保存，关闭即可。



```
from .core import InvalidData, DownloadFailure
from .platforms.debian import APT_INSTALLER
from .platforms.osx import BREW_INSTALLER
from .platforms.redhat import YUM_INSTALLER
from .rostdistrohelper import get_targets, get_release_file, PreRep137Warning

from .rep3 import download_targets_data # deprecated, will output warning

import warnings

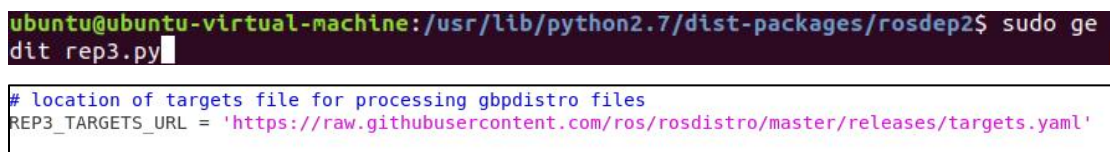
create_default_installer_context = None

# py3k
try:
    unicode
except NameError:
    basestring = unicode = str

# location of an example gbpdistro file for reference and testing
FUERTE_GBPDISTRO_URL = 'https://gitee.com/wybro/rostdistro/raw/' \
    'master/releases/fuerte.yaml'

# seconds to wait before aborting download of gbpdistro data
DOWNLOAD_TIMEOUT = 15.0
```

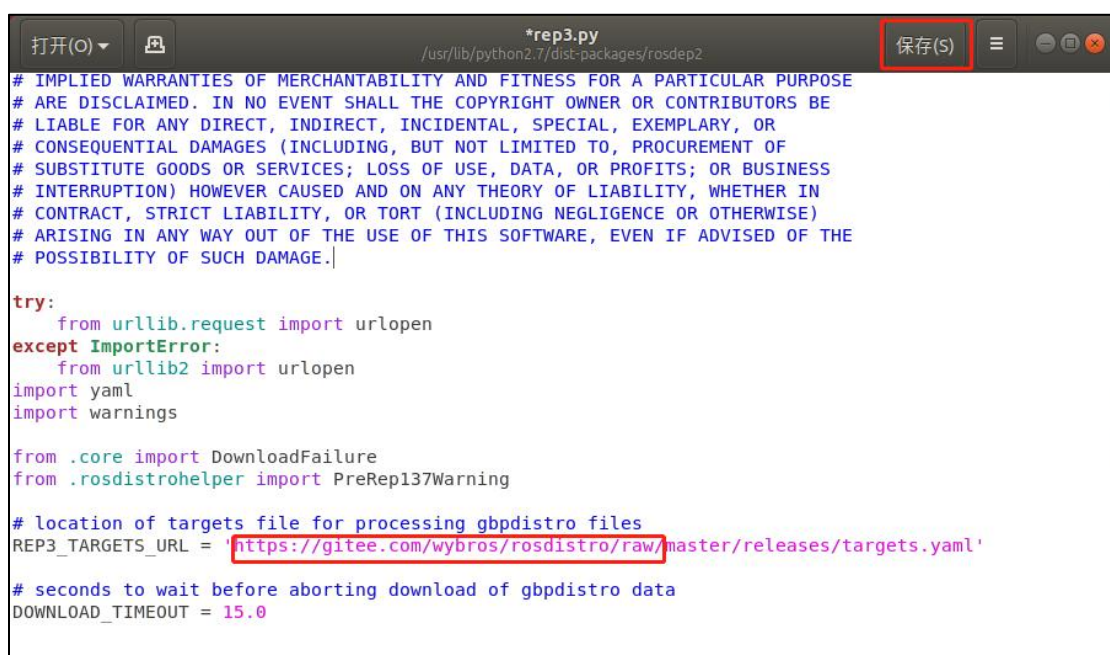
9) 接着输入指令“**sudo gedit rep3.py**”，打开“**rep3.py**”文件，找到“**REP3_TARGETS_URL**”代码。



```
ubuntu@ubuntu-virtual-machine:/usr/lib/python2.7/dist-packages/rosdep2$ sudo gedit rep3.py

# location of targets file for processing gbpdistro files
REP3_TARGETS_URL = 'https://raw.githubusercontent.com/ros/rosdistro/master/releases/targets.yaml'
```

10) 将“**https://raw.githubusercontent.com/ros/rosdistro/**”修改为“**https://gitee.com/wybro/rostdistro/raw/**”，（只修改下图红框所示内容,后续内容无需更改）修改完成后点击保存，关闭即可。



```
# IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE
# ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR CONTRIBUTORS BE
# LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR
# CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF
# SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS
# INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN
# CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE)
# ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE
# POSSIBILITY OF SUCH DAMAGE.

try:
    from urllib.request import urlopen
except ImportError:
    from urllib2 import urlopen
import yaml
import warnings

from .core import DownloadFailure
from .rostdistrohelper import PreRep137Warning

# location of targets file for processing gbpdistro files
REP3_TARGETS_URL = 'https://gitee.com/wybro/rostdistro/raw/master/releases/targets.yaml'

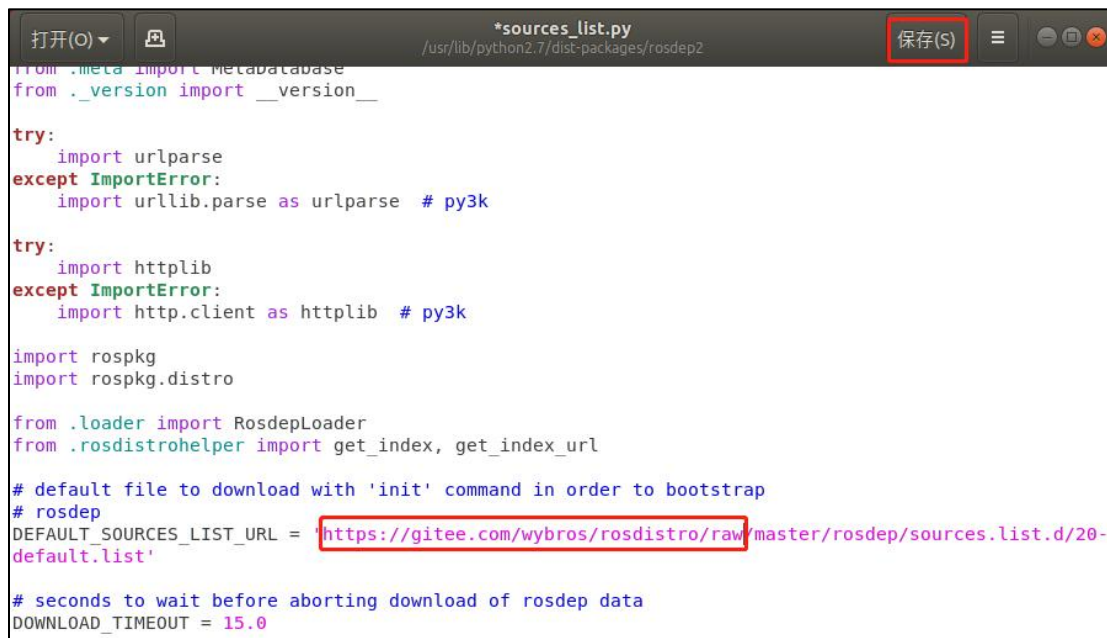
# seconds to wait before aborting download of gbpdistro data
DOWNLOAD_TIMEOUT = 15.0
```

11) 接着输入指令“`sudo gedit sources_list.py`”，打开“`sources_list.py`”文件，找到“`DEFAULT_SOURCES_LIST_URL`”代码。

```
ubuntu@ubuntu-virtual-machine:/usr/lib/python2.7/dist-packages/rosdep2$ sudo gedit sources_list.py
```

```
# default file to download with 'init' command in order to bootstrap
# rosdep
DEFAULT_SOURCES_LIST_URL = 'https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/sources.list.d/20-default.list'
```

12) 将“`https://raw.githubusercontent.com/ros/rosdistro/`”修改为“`https://gitee.com/wybro/srosdistro/raw/`”，（只修改下图红框所示内容,后续内容无需更改）修改完成后点击保存，关闭即可。



```
*sources_list.py
/usr/lib/python2.7/dist-packages/rosdep2
```

```
from .meta import Metadatabase
from ._version import __version__

try:
    import urlparse
except ImportError:
    import urllib.parse as urlparse # py3k

try:
    import httplib
except ImportError:
    import http.client as httplib # py3k

import rospkg
import rospkg.distro

from .loader import RosdepLoader
from .rosdistrohelper import get_index, get_index_url

# default file to download with 'init' command in order to bootstrap
# rosdep
DEFAULT_SOURCES_LIST_URL = 'https://gitee.com/wybro/srosdistro/raw/master/rosdep/sources.list.d/20-default.list'

# seconds to wait before aborting download of rosdep data
DOWNLOAD_TIMEOUT = 15.0
```

13) 输入指令“`cd /usr/lib/python2.7/dist-packages/rosdistro/`”，进入“`rosdistro`”目录下。

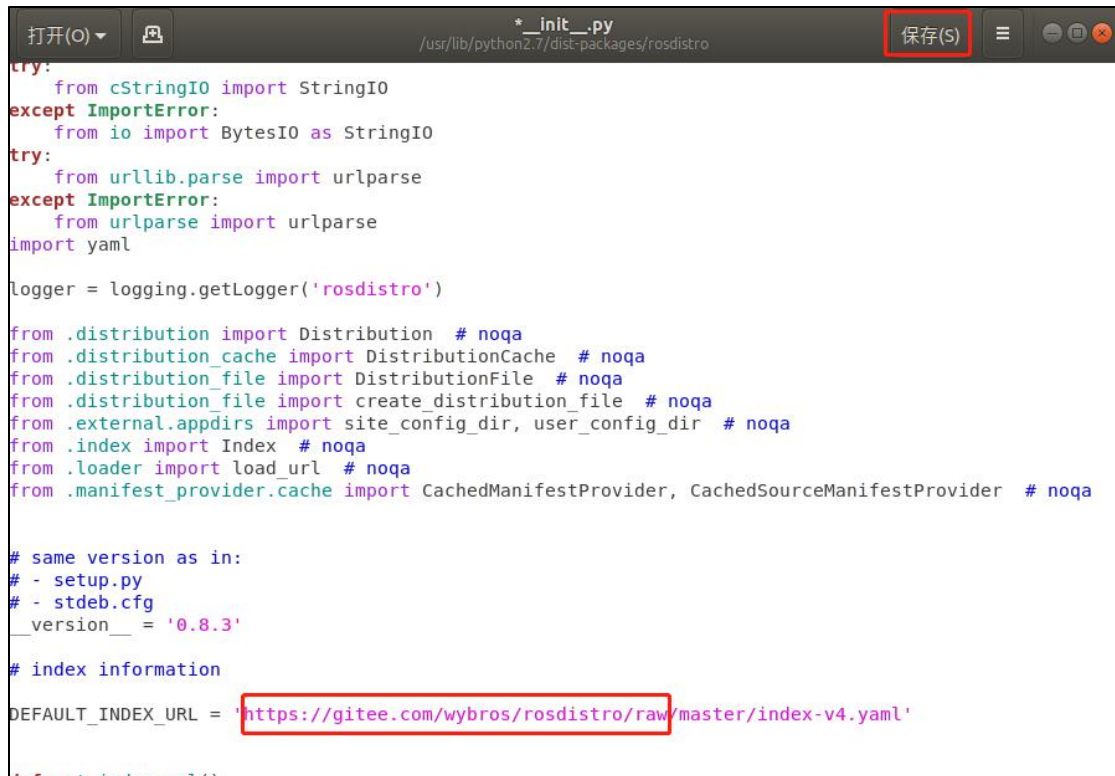
```
ubuntu@ubuntu-virtual-machine:/usr/lib/python2.7/dist-packages/rosdep2$ cd /usr/lib/python2.7/dist-packages/rosdistro/
```

14) 输入指令“`sudo gedit __init__.py`”，进入“`__init__.py`”文件，找到“`DEFAULT_INDEX_URL`”代码。

```
ubuntu@ubuntu-virtual-machine:/usr/lib/python2.7/dist-packages/rosdistro$ sudo gedit __init__.py
```

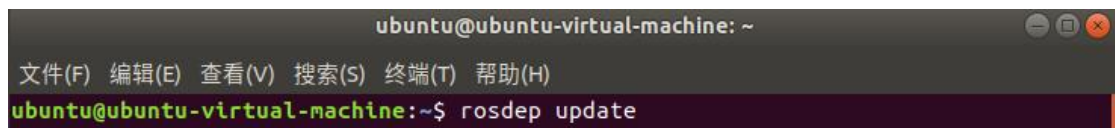
```
# index information
DEFAULT_INDEX_URL = 'https://raw.githubusercontent.com/ros/rosdistro/master/index-v4.yaml'
```

15) 将“<https://raw.githubusercontent.com/ros/rosdistro/>”修改为“<https://gitee.com/wybro/s/rosdistro/raw/>”，（只修改下图红框所示内容,后续内容无需更改）修改完成后点击保存，关闭即可。



8.更新

打开新的命令行终端，输入指令“**rosdep update**”更新。



出现如下信息，则代表更新成功了。

```
ubuntu@ubuntu-virtual-machine: ~  
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)  
Hit https://gitee.com/wybro/rostdistro/raw/master/rosdep/base.yaml  
Hit https://gitee.com/wybro/rostdistro/raw/master/rosdep/python.yaml  
Hit https://gitee.com/wybro/rostdistro/raw/master/rosdep/ruby.yaml  
Hit https://gitee.com/wybro/rostdistro/raw/master/releases/fuerte.yaml  
Query rostdistro index https://gitee.com/wybro/rostdistro/raw/master/index-v4.yaml  
Skip end-of-life distro "ardent"  
Skip end-of-life distro "bouncy"  
Skip end-of-life distro "crystal"  
Skip end-of-life distro "dashing"  
Skip end-of-life distro "eloquent"  
Add distro "foxy"  
Add distro "galactic"  
Skip end-of-life distro "groovy"  
Skip end-of-life distro "hydro"  
Skip end-of-life distro "indigo"  
Skip end-of-life distro "jade"  
Skip end-of-life distro "kinetic"  
Skip end-of-life distro "lunar"  
Add distro "melodic"  
Add distro "noetic"  
Add distro "rolling"  
updated cache in /home/ubuntu/.ros/rosdep/sources.cache  
ubuntu@ubuntu-virtual-machine:~$
```

若出现下图读取超时的提示，可能是网络原因，用户可多输入几次“**rosdep update**”尝试更新；若还是更新不了，用户可检查“[1.7 初始化](#)”的内容，确保修改的内容无误，后再次输入“**rosdep update**”进行更新。

```
ubuntu@ubuntu-virtual-machine:/usr/lib/python2.7/dist-packages/rostdistro$ rosdep update  
reading in sources list data from /etc/ros/rosdep/sources.list.d  
Query rostdistro index https://raw.githubusercontent.com/ros/rostdistro/master/index-v4.yaml  
Skip end-of-life distro "ardent"  
Skip end-of-life distro "bouncy"  
Skip end-of-life distro "crystal"  
Skip end-of-life distro "dashing"  
Skip end-of-life distro "eloquent"  
Add distro "foxy"  
Add distro "galactic"  
ERROR: error loading sources list:  
The read operation timed out (https://raw.githubusercontent.com/ros/rostdistro/master/galactic/distribution.yaml)
```